

Bits and Ops

A Complete Theory of Information

Registry: [\[HOWL-INFO-14-2026\]](#)

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Domain: Information Theory / Information Processing Theory

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1. Two Halves of One Subject

Information does two things. It moves between points, and it gets acted on at the points. These are not the same activity. Moving a message from New York to London is different from understanding it once it arrives. Transmitting a chest X-ray to a radiologist is different from the radiologist reading it. Delivering a notification to a developer's screen is different from the developer deciding what to do about it. Movement and action. Transit and processing. Channel and endpoint.

In 1948, Claude Shannon formalized the first half. His mathematical theory of communication gave exact answers to the questions of movement: how many bits can a channel carry, how should a message be encoded for efficient transmission, how can noise be overcome. His theory is among the most successful in the history of science, and its unit — the bit, one binary distinction — became the foundation of the information age.

Shannon explicitly excluded the second half. What happens at the endpoints — how a source decides what to transmit, how a destination makes sense of what it receives — he declared out of scope. The source produces symbols; the destination receives them. What either does with those symbols, Shannon left to others.

This paper presents the second half and shows how both halves compose into a single complete theory. Shannon gave us the bit — the unit of information in transit. The framework presented here gives us the **op** — the unit of information under action. Together, bits and ops cover everything information does.

2. Processing

Any system that must act on information is a processor. A CPU executing instructions is a processor. A surgeon deciding where to cut is a processor. A pilot classifying a radar contact is a processor. A physician diagnosing a patient is a processor. A developer debugging code, a teacher explaining a concept, a manager making a decision, an air traffic controller separating aircraft — each is a processor acting on information to produce a result.

The word carries no implication of silicon. It means any entity that transforms information into action, on any substrate. Biological processors run on neurons. Computational processors run on transistors. Organizational processors run on people coordinating. The substrate determines speed and capacity. The structure of processing is the same.

Processors share one constraint regardless of what they are made of. They operate on one element at a time. A CPU has one program counter — it executes one instruction, then the next. A surgeon has one pair of hands at one operative site. A pilot has one target in the gunsight. A human has one focus of conscious attention — one thought held in full clarity at a time. This is not a limitation to overcome through parallelism. Parallelism runs multiple processors, each of which still operates on one element. The constraint is the definition of what an operation is: a single transformation applied to a single element by a single processor.

3. The Op

The unit of processing is the op: one irreducible transformation by one processor. A diagnostic question asked by a physician. A mirror glance by a driver. A cache lookup by a CPU. A line of code read and understood by a developer. A comparison between two elements by a sorting algorithm. A decision to turn left or right by a pilot. Each is one op.

The op is to processing what the bit is to transmission. Shannon showed that a bit over fiber optic and a bit over copper wire and a bit by postal mail are the same bit. The medium determines how fast bits travel, not what a bit is. Similarly, an op by a CPU and an op by a surgeon and an op by a pilot are the same op. The substrate determines how long an op takes — nanoseconds for a CPU, seconds for a surgeon, hours for a strategic planner — but an op is an op. It is countable. It is observable. It is universal across every domain that processes information, which is every domain.

If a physician executes fourteen ops to reach a diagnosis, their processing cost for that diagnosis is fourteen. If an experienced physician has internalized the pattern so thoroughly that the diagnosis fires without conscious effort, their processing cost is zero. The cost is not abstract. It is the number you get when you count the operations.

4. The Time Budget

Processing is bounded by one inequality:

$$\text{Total ops} \times \text{average op duration} \leq \text{time budget}$$

The time budget is fixed by domain physics. For a driver, it is the lane tolerance — the fraction of a second before the car drifts into the guardrail. For a surgeon, it is the anesthesia window — the hours before patient risk escalates. For a web server, it is the request timeout — the milliseconds before the client disconnects. For a fighter pilot, it is the engagement window — the seconds before the adversary shoots. The time budget is the wall. It does not negotiate. It does not extend.

Average op duration is how long each op takes for this processor in this context. A CPU op takes nanoseconds. A surgical op takes seconds. A strategic planning op takes hours. You generally cannot make ops faster — you cannot speed up the surgeon's hands or the CPU's clock.

The only lever is reducing the total number of ops. This single fact drives everything that follows in this paper. Every mechanism, every concept, every strategy described from this point forward is, in one way or another, a way to keep the left side of this inequality below the right. Reduce the ops. The budget is the wall.

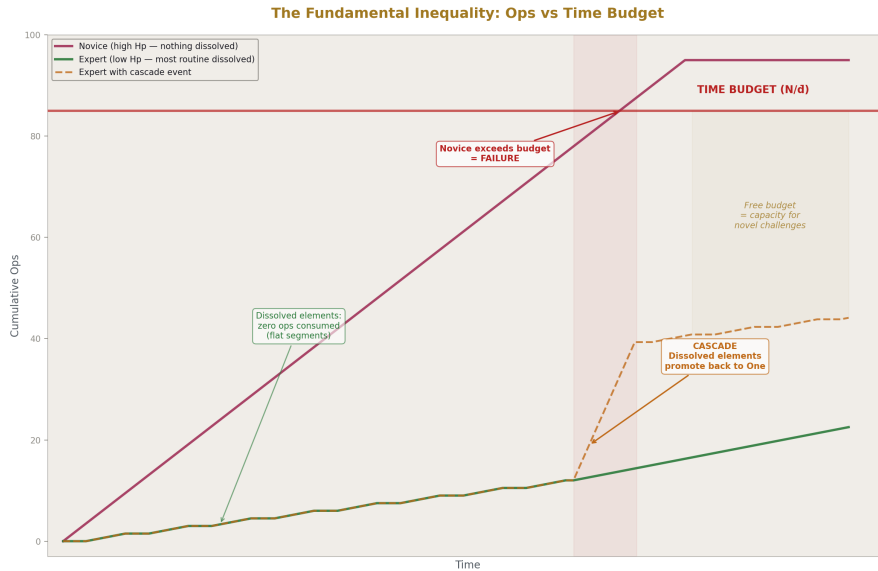


Figure 1: Fig. 5: Budget Consumption — novice hits the wall while expert has free capacity; cascade event shows temporary spike toward budget ceiling.

5. Three Cardinalities

Every element that a processor might need to act on exists, relative to that processor, at one of three cardinalities. These are not categories assigned by a designer. They are intrinsic properties of the relationship between the element and the processor.

Infinity. A population. Multiple elements present, none selected, the processor unable to act on the population as a population. The forty symptoms a physician must evaluate before reaching a diagnosis. The million rows a database must scan. The twelve aircraft on the air traffic controller's radar. The twenty tasks on the developer's backlog. Until this population is reduced to one specific element that the processor acts on, no work occurs. The processor is stalled, regardless of how powerful it is.

One. The single element under active operation. The only cardinality at which work happens. The one patient on the operating table. The one bug being debugged. The one target being engaged. The one sentence being spoken. Everything in processing exists to reach this state and act here.

Zero. Outside the processor's operational boundary. The processor can observe but cannot act. Weather for a farmer — measurable, completely uncontrollable. The speed of light for a network engineer — a permanent physical constraint that defines minimum latency. Hardware failure rates — they happen on physics' schedule, not the system administrator's. The BIOS firmware that executed before the operating system existed — the OS references those events but cannot change them. Zero elements emit events into the system but cannot receive operations back. They define the boundary of what the processor can influence.

The interaction between cardinalities determines what is possible. One interacting with Infinity is always fan-out (one orchestrator acting on many members in sequence) or convergence (many members reporting to one coordinator). Infinity interacting with Infinity is nested iteration — matching members from two populations. Zero interacting with anything is one-way: events flow from Zero into the system, never back. These patterns are not designed. They are mechanical consequences of the cardinalities involved. Declare “there is one scheduler” and “there are many processes” and the interaction pattern — the scheduler fans out to processes, processes converge back to the scheduler — follows automatically.

6. Reduction

Infinity must become One before work occurs. The mechanism that accomplishes this is a four-stage pipeline that appears identically across every domain.

Enumerate. Make the population explicit and finite. The physician takes a history and assembles a list of possible diagnoses. The CPU scheduler populates the run queue. The pilot registers all contacts on the radar scope. Before enumeration, the population is unknown — unbounded, formless, impossible to process. After enumeration, it is a finite list. Every subsequent stage operates on this list.

This is the most dangerous stage at which to fail. If the correct element is not enumerated — if the physician doesn't consider the right diagnosis, if the scheduler doesn't see the ready process, if the pilot doesn't register the contact — then no subsequent stage can recover. The processor will confidently act on the wrong element and never know, because the right answer was never in the candidate set. Enumeration failure is invisible from inside the pipeline.

Filter. Eliminate candidates that don't meet relevance criteria. The physician rules out diagnoses incompatible with the presentation. The scheduler excludes blocked processes. The pilot dismisses contacts identified as friendly. The population shrinks. Filtering can fail by over-filtering (the correct element was eliminated too early) or under-filtering (too many candidates remain and the next stage is overwhelmed — analysis paralysis).

Score. Evaluate remaining candidates against weighted considerations. The physician weighs symptom fit, disease prevalence, and test results against each remaining diagnosis. The scheduler weighs priority, time-since-last-run, and interactivity against each process. The pilot weighs proximity, threat level, and weapons capability against each contact. Candidates are ranked. Scoring can fail when the weights are wrong or a consideration is missing — the correct element is present but outscored by an inferior candidate.

Select. Commit the highest-scored candidate as One. The physician commits to a working diagnosis. The scheduler dispatches a process to the CPU. The pilot commits to an engagement. Infinity has become One. Work proceeds. Selection can fail when two candidates score nearly equally and the processor oscillates — unable to commit despite having all information. In time-critical domains, this oscillation consumes the time budget while producing nothing.

The pipeline itself has a cost. Enumerating, filtering, scoring, and selecting all consume ops — ops that could have been spent acting. The CPU scheduler runs on CPU cycles. The pilot's attention spent classifying threats is attention not spent maneuvering. The physician's time weighing differentials is time not spent treating. Reduction consumes the resource it allocates. The optimal reduction is the minimum correct steps to reach an actionable One — not the most thorough analysis, but the most efficient one that still produces a correct result.

7. Dissolution

Through repetition under consistent conditions, a processing chain's op count decreases toward zero. This is the framework's central mechanism and the phenomenon that explains expertise, caching, compilation, muscle memory, habit, pattern recognition, and fluency across every domain.

A new driver checks the rear-view mirror in six operations: locate the mirror, focus on it, scan the reflected image, identify objects in the image, assess whether any object is a threat, return gaze to the road. After six months of daily driving, the same check takes two operations. After five years, it takes zero. The check still happens — the driver is

not blind to what's behind them — but it happens structurally, without consuming the conscious pipeline. The processing has dissolved into the substrate.

Dissolution is not forgetting. Forgetting is loss — the capability is gone. Dissolution is compression into structure — the capability produces correct results without consuming the scarce resource. The experienced physician who diagnoses pneumonia at a glance has not forgotten the reasoning chain. The chain fires structurally, instantly, at zero pipeline cost. Every op of reasoning that a student would perform consciously, the expert performs structurally.

A CPU cache is dissolution. A memory access that once cost two hundred cycles (fetching from main memory) now costs four cycles (retrieving from cache). The chain was executed once, the result stored, and now the result is produced structurally. A branch predictor is dissolution. A compiled routine is dissolution. An expert's pattern recognition is dissolution. A factory worker's muscle memory is dissolution. A fluent speaker's vocabulary is dissolution. The mechanism is universal: repetition under consistent conditions converts active processing into structural processing that is free.

Dissolution transforms the fundamental inequality. Elements that have dissolved cost zero ops. They don't count against the time budget. The more elements dissolved, the more budget is available for novel challenges. This is what expertise is — not knowing more facts but having dissolved more processing chains into structure, freeing the pipeline for the genuinely new. The expert surgeon's hands are not faster than the novice's. The expert performs fewer operations because most of the processing that the novice must do consciously has dissolved into structure.

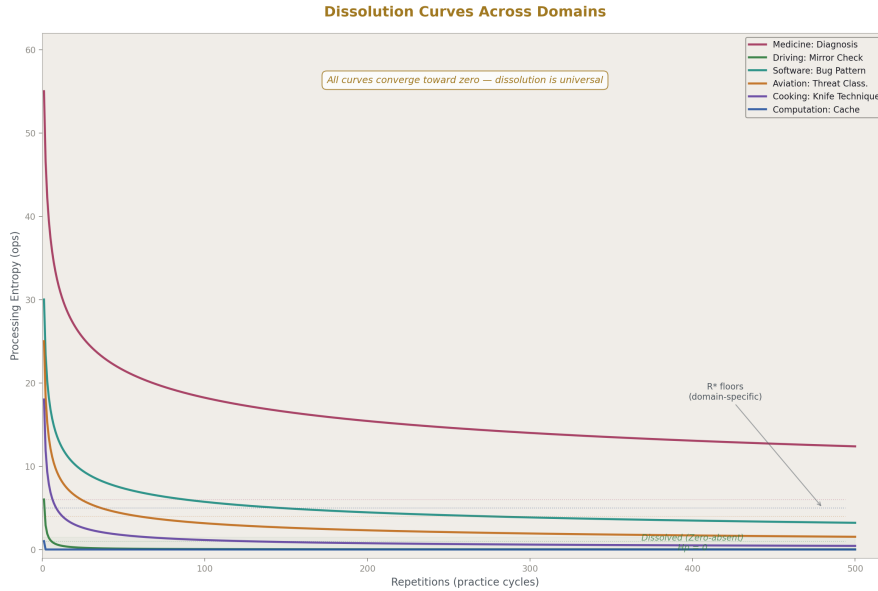


Figure 2: Fig. 1: Dissolution Curves Across Domains — all curves converge toward zero despite different starting points and rates, showing dissolution is universal.

8. Four States

Combining cardinality and dissolution produces four states that every element occupies relative to a processor, a goal, and a context. The states are mutually exclusive and exhaustive. Nothing exists outside them.

Infinity. Population awaiting reduction. Multiple candidates, none selected. The unsorted array. The undiagnosed patient's symptom list. The unread emails. Processing entropy is positive — each element will cost ops when promoted to One.

One. Under active operation. The element currently being processed. The sort comparison being made. The diagnosis being reasoned through. The email being read. Processing entropy is positive and being consumed in real time.

Zero-absent. Dissolved into structure. This element was once at One. It was processed repeatedly under consistent conditions. The processing chain collapsed. Now the correct result is produced structurally at zero op cost. The native-language word that decompresses without effort. The familiar route driven without conscious navigation. The cached memory value retrieved at L1 speed. The expert's diagnostic pattern that fires instantly. Processing entropy is zero — not because processing isn't happening, but because it happens structurally, without pipeline allocation.

Zero-external. Permanently outside the processor's domain. This element was never at One and never will be. The weather for the farmer. The speed of light for the network engineer. Hardware degradation physics. Biological aging. Other drivers' decisions. The processor cannot operate on these elements at any op cost. Processing entropy is undefined — not zero (which means dissolved) and not high (which means expensive but possible). The processor simply cannot act here. The correct response is not processing but structural resilience: build systems that survive the boundary's effects without attempting to change them.

State is a property of the relationship among element, processor, goal, and context — not a property of the element alone. The same chest X-ray is at Zero-absent for the experienced radiologist (dissolved, instant diagnosis), at One for the radiology resident (under active interpretation), at Infinity for the emergency physician who has twenty films to review (population awaiting reduction), and at Zero-external for the hospital's billing system (cannot process medical images at all). Same element. Four different states. Four different processors.

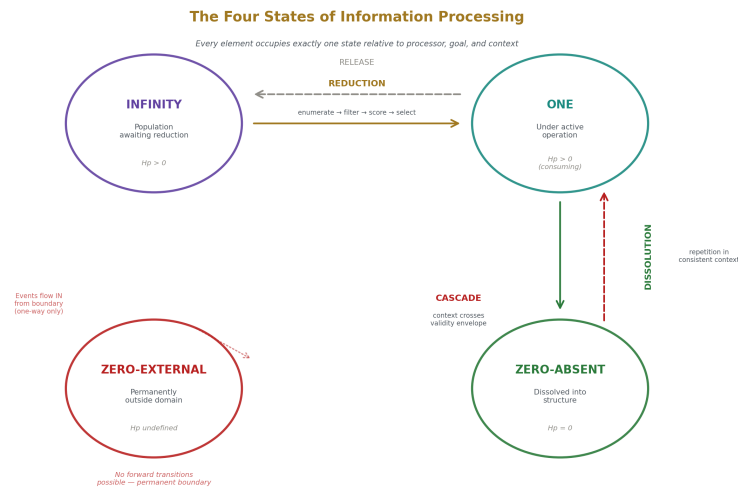


Figure 3: Fig. 2: The Four States — Infinity, One, Zero-absent, Zero-external with allowed transitions and cascade regression path.

9. Dissolution Has Conditions

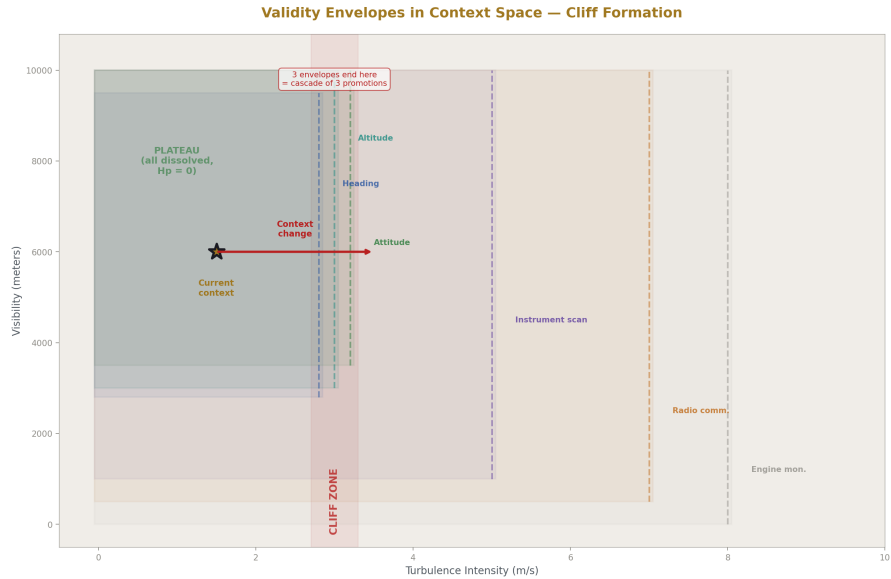


Figure 4: Fig. 4: Validity Envelopes in 2D Context Space — overlapping envelopes with cliff zone where boundaries align, causing simultaneous cascade.

Every dissolved element dissolved under specific conditions. The pilot's altitude maintenance dissolved under visual flight in calm air in a familiar aircraft. The surgeon's knot-tying dissolved under normal tissue tension with standard suture material. The CPU's cache entry dissolved under a specific memory access pattern with a specific working set size. These conditions define a region — the **validity envelope** — within which the dissolution holds.

Inside the envelope, the element costs zero ops. Outside the envelope, it promotes back to One and costs ops again. The envelope is not a metaphor. It is a definable region in the space of operating conditions, with measurable boundaries along each dimension: visibility range, turbulence level, aircraft type, physiological state for the pilot; tissue characteristics, instrument availability, team composition for the surgeon; access pattern locality, working set size, competing process interference for the cache.

Envelope width is determined by dissolution conditions. An element dissolved under narrow conditions — always the same aircraft, always clear weather, always daylight — has a narrow envelope. An element dissolved under varied conditions — multiple aircraft types, weather ranging from clear to marginal, day and night — has a wide envelope. You cannot dissolve wider than you have practiced. Training under varied conditions produces wider envelopes. This is the formal explanation for why varied practice produces more robust expertise than repetitive drilling under identical conditions.

When a context change pushes operating conditions outside the validity envelope of a dissolved element, that element promotes back to One. It costs ops again. When a context change crosses many envelopes simultaneously, many elements promote at once. This is a **cascade**: a sudden spike in total ops that may overwhelm the pipeline's capacity.

A bee enters the cockpit. A tiny event. The pilot swats at it, loses visual reference for two seconds. Three dissolved tasks promote simultaneously: altitude maintenance, heading maintenance, attitude awareness. Each now requires conscious processing. The pipeline handles one at a time. Three compete. Performance degrades catastrophically — not because the pilot forgot anything, but because the context crossed three validity envelopes at once, and the pipeline that was comfortably handling one conscious task now has four.

Cascade severity is the count of simultaneous promotions. It is completely independent of the trigger's magnitude. The bee is tiny. The cascade is large. A thunderclap — a much larger event — that only invalidates one dissolution is much less severe than the bee. Severity is a property of the fragility topology, not the triggering event. This is why small disruptions sometimes cause catastrophic failures while large ones are absorbed: the small disruption happened to cross a cluster of validity envelope boundaries that the large one didn't.

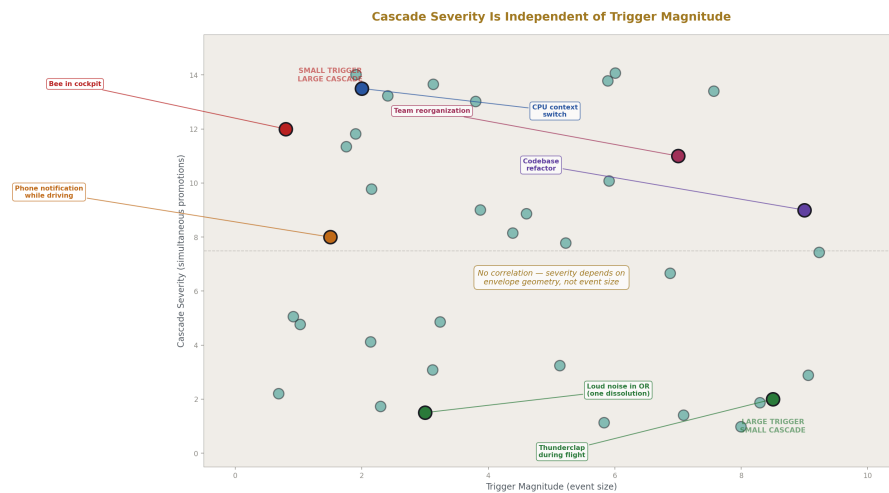


Figure 5: Fig. 3: Cascade Severity vs Trigger Magnitude — no correlation; severity depends on envelope geometry, not event size.

10. Processing Entropy

Processing entropy is the op count a specific processor requires for a specific element, given a specific goal in a specific context. It is the measure of endpoint cost — what Shannon excluded.

$H_p(x \mid p, g, c)$ = ops for processor p to reduce element x to actionable One for goal g in context c

Processing entropy is zero when the element is at Zero-absent: dissolved, no ops required, the result produced structurally. It is positive when the element is at Infinity or One: ops required, pipeline consumed, budget spent. It is undefined when the element is at Zero-external: the processor cannot act on this element at any cost.

Processing entropy is receiver-dependent. This is its defining distinction from Shannon's entropy, which is a property of the source regardless of who receives the message. Shannon's entropy for a chest X-ray is determined by the image's statistical properties — the same number regardless of who views it. Processing entropy for the same X-ray is three ops for the experienced radiologist (dissolved pattern recognition) and twenty-eight ops for the first-year resident (methodical systematic review). Same image. Same Shannon entropy. Different processing entropy. The difference is dissolution state.

Processing entropy collected across a set of tasks for one processor produces a **profile** — a vector showing the cost landscape across the processor's operational domain. The expert's profile is mostly zeros with a few positive peaks. The novice's profile is mostly high values with few zeros. The distance between two profiles is a measurable quantity, giving expertise a geometry: skill gaps become distances with direction, learning becomes a trajectory through a metric space, and clusters of similar practitioners become formally identifiable objects.

Maturity, for any processor, is the systematic reduction of processing entropy toward zero across the operational domain. Each dissolution converts one element's processing entropy from positive to zero. The progression from novice to expert is visible in profile space as a trajectory from far-from-origin (everything is expensive) to near-origin (most things are free). The trajectory's shape, speed, and direction are measurable.

11. Optimal Reduction

Every task has a floor: the minimum number of correct ops any competent processor requires for reliable execution. This is R^* — the optimal reduction. Below R , *the processor is not dissolved — it is operating without sufficient verification, which means it is guessing*. Above R , there is measurable inefficiency that practice can absorb. Between R^* and zero is the space that dissolution fills — converting minimum-competent processing into structural processing that costs nothing.

R^* connects to Shannon through a precise parallel. Shannon's source coding theorem says: encode at the entropy rate, no fewer bits. Below the entropy rate, information is lost. Above it, bits are wasted. Optimal reduction says: reduce to actionability, no fewer ops. Below R , *correctness is lost*. Above R , ops are wasted. Both are optimality conditions defining a floor below which quality degrades and above which resources are squandered.

Not all floors are equally knowable. Sorting N elements requires at least $N \log N$ comparisons — a provable mathematical theorem derived from the information structure of the

task. Nobody can beat it because the proof demonstrates impossibility. An emergency physician diagnosing chest pain takes five to eight ops at expert level — an empirical observation that nobody can prove is optimal. Maybe four ops suffice. Maybe three. Nobody knows, because the task lacks the formal structure that would enable a proof.

This difference — whether R^* is provable from the task's structure or only estimable from expert observation — classifies every task into one of three derivability classes. Provable: R^* is a theorem (sorting, searching, parity checking). Boundable: R^* is constrained between proven limits that don't meet (NP-hard optimization, some diagnostic criteria). Empirical: R^* is known only from the best observed performance (most medical diagnosis, tactical combat, creative problem-solving, novel debugging). The class determines what training can target, what assessment can measure, and what dissolution infrastructure can guarantee.

12. The Concurrency Tax

A function that takes 4 milliseconds in a benchmark takes 23 milliseconds in production. The function is the same. The processor is the same. What changed is the environment: other activities are competing for shared resources. The overhead has five components.

Contention. The resource you need is held by someone else. The database connection is occupied. The instrument is in the assistant's hand. The CI pipeline is running someone else's build. You wait. Your op duration inflates by the waiting time. The op itself is unchanged — same transformation — but the calendar time is longer.

Cascade. Another stream's activity breaks your dissolved processing. A context switch evicts your cache entries: each subsequent access that was a four-cycle dissolved hit becomes a two-hundred-cycle active fetch. An interruption evicts the developer's mental model from working memory: fifteen to twenty-five minutes of re-orientation to reconstruct what was free moments ago. Cascade adds ops that would not exist if you were working in isolation.

Coordination. Ops spent managing concurrent access. Lock acquisition. Handoff protocols. Meeting agendas. Status updates. Merge conflict resolution. These ops advance no task. They exist purely because concurrency exists.

Blocking. The pipeline is idle. Not inflated-duration busy — completely idle. Waiting for a resource that is occupied and critical to your path. A thread sleeping on a lock. A surgeon waiting for a pathology result. A developer waiting for code review approval. Budget consumed at zero work output.

Interleave. Ops spent deciding which stream to service. The scheduler choosing the next process. The surgeon deciding whether to address the nurse's question or finish the suture. The developer triaging between the Slack message, the code review, and the bug. Each decision is an op that advances no stream.

The concurrency tax is not random. It is derivable from the **contention graph** — the topology of shared resources and competing streams. Resources are nodes. Streams con-

nect to the resources they need. Two streams sharing many resources contend frequently. A resource connected to many streams is a bottleneck. The topology determines the scaling law: star topologies (one central resource, all streams competing) have steep, divergent tax growth. Partitioned topologies (groups sharing within, independent between) have bounded tax. Hierarchical topologies (cheap local resources, expensive global ones) have logarithmic common-case tax.

Expertise reduces the tax. Experts have dissolved the interleave decisions (structural response to interruptions rather than deliberative evaluation), widened validity envelopes (reduced cascade exposure — the expert’s dissolved processing survives interruptions that break the novice’s), and dissolved coordination protocols (a glance replacing a sentence, a gesture replacing an instruction). The expert’s concurrency tax on the same contention graph is measurably lower.

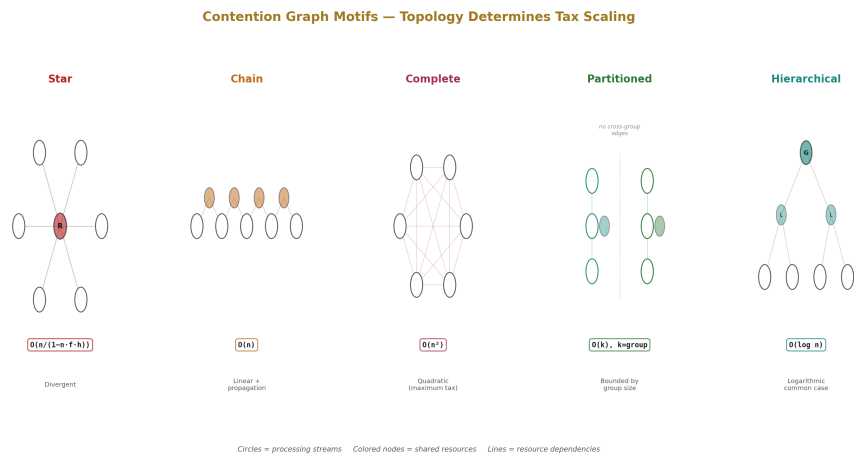


Figure 6: Fig. 8: Five Contention Graph Motifs — star, chain, complete, partitioned, hierarchical with characteristic scaling laws.

13. Compression and Language

A word is a compression token. It packs many referents into one transmissible symbol. “Fire” compresses combustion, employment termination, weapon discharge, ceramic kiln process, and dozens of other meanings into a single syllable. “Eigenvalue” compresses an entire mathematical concept into one word. Compression is how language achieves its bandwidth.

Decompression is context-dependent. “Fire” in a burning building means combustion. “Fire” in a boardroom means termination. The receiver resolves the ambiguity by applying dissolved context-dependent decompression rules — rules that map (token, context) to the

intended referent. When the rules are dissolved, decompression is free. When they are not dissolved, each unfamiliar token costs ops to decode.

Language is a shared codebook dissolved to zero decompression cost across a population. Fluent conversation is possible only because common words cost zero ops at both ends. If every word required conscious decoding, normal speech — a hundred fifty words per minute — would overwhelm the pipeline before the first sentence finished. Fluency is dissolved decompression. Every word that is not dissolved is a processing tax on the message.

The compression ratio — referents per token — grows with experience. A child’s “fire” decompresses to two or three referents. An adult’s to fifteen. A firefighter’s to fifty. An arson investigator’s to a hundred. Experience dissolves more referent-to-token associations, expanding what each symbol can carry. Civilization itself is the accumulated stack of compression tokens dissolved across populations. The alphabet dissolved thousands of logograms into twenty-six composable letters. Positional notation dissolved per-scale arithmetic. Standardized containers dissolved per-cargo handling. Each innovation created a compression token or dissolution infrastructure, dissolved it across a population, and freed processing capacity for the next unsolved problem.

14. The Three-Term Cost

Every act of communication has three costs:

$$\text{Cost}(A \rightarrow B) = \text{Hp}(A, \text{encode}) + \text{Hs}(\text{channel}) + \text{Hp}(B, \text{decode})$$

Sender ops to transform thought into transmissible symbols. An expert explaining a dissolved concept: near zero — the words come without effort. A novice explaining something they barely understand: high — they must actively search for words, simplify, reformulate.

Channel bits to move the symbols from sender to receiver. Shannon’s domain, fully formalized. The bits required for reliable transmission. This term depends on the message and the channel. It does not depend on who is sending or receiving.

Receiver ops to transform received symbols into understanding. An experienced physician reading a radiology report: one or two ops — terminology dissolved, structure dissolved, implications automatic. A medical student reading the same report: twenty to thirty ops — each term consciously decoded, the structure actively parsed, the implications explicitly constructed.

Shannon formalized the middle term and gave it exact mathematics. His framework is recovered as the special case where both endpoint terms are zero — when both sender and receiver have dissolved the relevant codebook. Expert-to-expert communication in a shared domain approaches this case. Two experienced surgeons discussing a familiar procedure. Two senior developers discussing a shared codebase. The words are free at both ends. The only cost is the channel.

For most real-world communication, the endpoint terms dominate. Modern channels are cheap — email costs fractions of a cent, web pages load in milliseconds. Processing is expensive — the junior developer spending forty-five minutes wrestling with an unfamiliar API is paying in ops, not bits. Shannon optimized the cheapest term. Processing entropy theory formalizes the expensive ones.

The **dissolution differential** — the gap between sender and receiver processing entropy for the same tokens — predicts communication difficulty. When the expert uses terminology that costs them zero and costs the receiver ten ops per term, the message arrives perfectly on the channel and fails at the endpoint. Adding explanatory redundancy (definitions, examples, context) increases channel cost to reduce receiver processing cost. Shannon-optimal encoding minimizes channel bits. Processing-optimal encoding minimizes total cost. They differ whenever the receiver’s processing entropy is high.

When one message serves many receivers with different dissolution states, no single encoding optimizes for all. The expert wants compressed shorthand. The novice needs expanded explanation. The optimal strategy is layered encoding — compressed base content with expandable dissolution infrastructure. Documentation, teaching, API design, and user interface design are all instances of this optimization: minimizing total cost across a heterogeneous audience by providing the right amount of dissolution infrastructure at each layer.

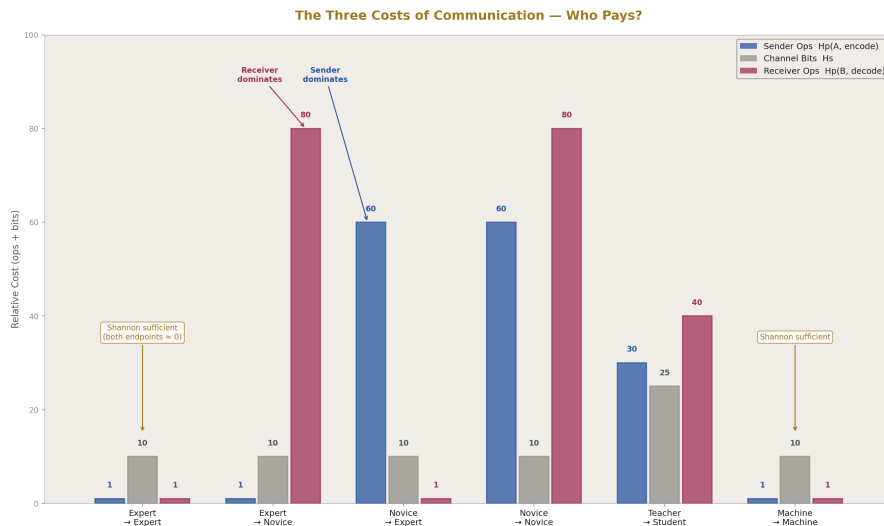


Figure 7: Fig. 6: Three-Term Cost Scenarios — sender ops, channel bits, and receiver ops across six scenarios showing which term dominates.

15. Bits and Ops

Information does two things. It moves and it gets acted on.

Shannon gave us the mathematics of movement. The bit — one binary distinction — is the unit. Channel capacity bounds throughput. Entropy rate determines minimum encoding. Error correction overcomes noise. His framework is complete for the middle of the communication chain: source encoding, channel transmission, destination decoding. Sixty years of engineering built on it. The digital world runs on it.

This paper presents the mathematics of action. The op — one irreducible transformation — is the unit. The time budget bounds throughput. Optimal reduction determines minimum processing. Dissolution overcomes cost. The framework is complete for the endpoints Shannon excluded: what the source does before encoding and what the destination does after decoding.

The two halves compose. Total cost of any information activity: ops to encode, bits to transmit, ops to decode. Three terms, two units, one equation. Shannon's theory is not superseded — it is completed. His framework remains exact for the channel. This framework covers the territory he deliberately left blank.

Processing is bounded by one inequality: total ops times op duration cannot exceed the time budget. Three cardinalities are intrinsic: Zero is the boundary, One is where work happens, Infinity is the population awaiting reduction. Four states emerge: Infinity pending, One active, Zero-absent dissolved, Zero-external permanent boundary. Reduction is the four-stage pipeline from Infinity to One. Dissolution is the mechanism that moves elements from One to Zero-absent, converting active processing to structural processing that is free. Cascades are what happens when dissolution breaks. Processing entropy measures the cost. Optimal reduction defines the floor. The concurrency tax explains why things cost more in practice. Compression tokens give language its bandwidth. The three-term equation unifies transmission and processing into one cost.

Bits for what moves. Ops for what happens. Together, the complete theory of information.

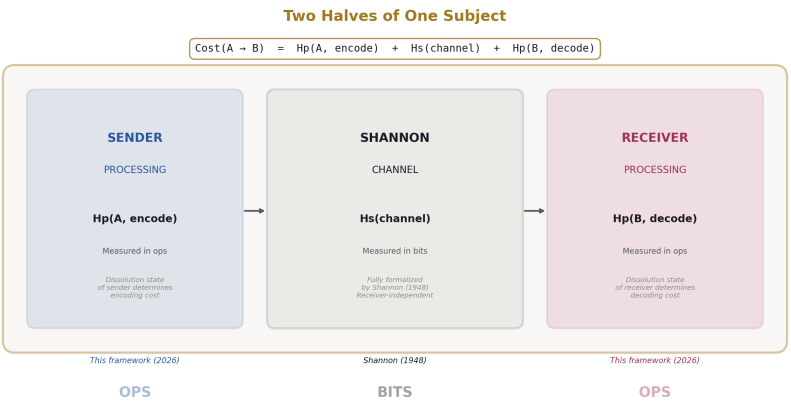


Figure 8: Fig. 7: Shannon’s Scope vs Complete Framework — Shannon formalized the middle third; this framework covers the endpoints he excluded.

Appendix: Supporting Tables

HOWL-INFO-14-2026

Table A: Complete Concept Inventory

Order	Concept	Definition	Unit	Introduced In Paper	Section in This Paper
1	Processor	Any system that transforms information into action; operates on one element at a time	—	HOWL-COMP-1-2026	2

Order	Concept	Definition	Unit	Introduced In Paper	Section in This Paper
2	Op	One irreducible transformation by one processor; universal unit of processing	count	HOWL-MATH-15-2026	3
3	Time budget (N)	Maximum duration available for processing; fixed by domain physics	time units	HOWL-MATH-15-2026	4
4	Fundamental inequality	$\bar{d} \times \text{ops} \leq N$; total processing bounded by time budget	ops $\times$ time $\leq$ time	HOWL-MATH-15-2026	4
5	Cardinality: Infinity	Population of elements; must be reduced to One before work occurs	—	HOWL-INFO-11-2026	5
6	Cardinality: One	Single element under operation; only state where work happens	—	HOWL-INFO-11-2026	5

Order	Concept	Definition	Unit	Introduced In Paper	Section in This Paper
7	Cardinality:Zero	Outside operational boundary; can observe, cannot act	—	HOWL-INFO-11-2026	5
8	Reduction pipeline	Four-stage mechanism (enumerate→filter→score→select) collapsing Infinity to One	ops consumed	HOWL-INFO-12-2026	6
9	Enumeration	Making population explicit and finite; unknown $N \rightarrow$ known N	—	HOWL-INFO-12-2026	6
10	Filtering	Eliminating irrelevant candidates; $N \rightarrow$ smaller N	—	HOWL-INFO-12-2026	6
11	Scoring	Evaluating candidates against weighted considerations; imposing rank	—	HOWL-INFO-12-2026	6

Order	Concept	Definition	Unit	Introduced In Paper	Section in This Paper
12	Selection	Committing highest-scored candidate as One; transition to action	—	HOWL-INFO-12-2026	6
13	Dissolution	Processing chain collapsing into structure through repetition; op count $\rightarrow$ zero	ops eliminated	HOWL-INFO-13-2026	7
14	State: Infinity	Population awaiting reduction; processing entropy positive	—	HOWL-INFO-13-2026	8
15	State: One	Under active operation; processing entropy positive and being consumed	—	HOWL-INFO-13-2026	8
16	State: Zero-absent	Dissolved into structure; was once at One; processing entropy = 0	—	HOWL-INFO-13-2026	8

Order	Concept	Definition	Unit	Introduced In Paper	Section in This Paper
17	State: Zero-external	Permanently outside domain; never was at One; processing entropy undefined	—	HOWL-INFO-13-2026	8
18	Validity envelope	Region of context space within which dissolution holds	region in context space	HOWL-MATH-16-2026	9
19	Cascade	Many dissolved elements promoting to One simultaneously when context crosses envelopes	count (promotions)	HOWL-MATH-16-2026	9
20	Processing entropy (Hp)	Op count for specific processor on specific element for specific goal in specific context	ops	HOWL-INFO-13-2026	10

Order	Concept	Definition	Unit	Introduced In Paper	Section in This Paper
21	Optimal reduction (R^*)	Minimum correct ops any competent processor requires	ops	HOWL-MATH-15-2026	11
22	Derivability classes	Classification of tasks by whether R^* is provable, boundable, or empirical	—	HOWL-MATH-20-2026	11
23	Concurrency tax	Additional ops from execution environment: contention + cascade + coordination + blocking + interleave	ops + time	HOWL-MATH-18-2026	12
24	Contention graph	Graph of shared resources and competing streams determining tax structure	—	HOWL-MATH-18-2026	12
25	Compression token	Symbol packing many referents into one transmissible unit	—	HOWL-INFO-13-2026	13

Order	Concept	Definition	Unit	Introduced In Paper	Section in This Paper
26	Compression ratio	Referents per token for a given processor; grows with experience	count	HOWL-INFO-13-2026	13
27	Three-term cost	$\text{Cost}(A \rightarrow B) = \text{Hp}(A, \text{encode}) + \text{Hs}(\text{channel}) + \text{Hp}(B, \text{decode})$	ops + bits + ops	HOWL-INFO-13-2026	14
28	Dissolution differential	Gap in processing entropy between sender and receiver for same tokens	ops	HOWL-MATH-19-2026	14
29	Layered encoding	Message structured in layers of increasing dissolution infrastructure	—	HOWL-MATH-19-2026	14
30	Bit	Shannon's unit: one binary distinction; unit of information in transit	bit	Shannon 1948	15

Table B: The Two Units

Property	Bit (Shannon)	Op (This Framework)
Measures	Information in transit	Information under action
Domain	Channel between processors	Endpoint at processor
Defined by	One binary distinction	One irreducible transformation
Depends on	Source statistics; channel properties	Processor dissolution state; element; goal; context
Receiver-dependent?	No — same bits regardless of receiver	Yes — same element, different ops for different processors
Zero means	No information (trivial source)	Dissolved — processing happens structurally at zero cost
Optimization principle	Encode at entropy rate, no fewer bits	Reduce to actionability, no fewer ops
Over-optimization penalty	Information loss (under-encoding)	Actionability loss (over-reduction)
Fundamental limit	Channel capacity C (bits/second)	Time budget $N / \bar{d}$ (ops/period)
Universal across	All channels regardless of physical medium	All processors regardless of substrate
Historical origin	Shannon, 1948	This series, 2026
Fungibility	Bit over fiber = bit over copper = bit by mail	Op by CPU = op by surgeon = op by pilot

Table C: Three Cardinalities

Cardinality	Nature	Can Emit Events	Can Receive Operations	Population Visibility	Computational Role	Example
Zero	Outside operational boundary	Yes (one-way into system)	No	N/A	Boundary; defines system character; source of initial events	Weather; speed of light; BIOS; hardware failure physics

Cardinality	Nature	Can Emit Events	Can Receive Operations	Population Visibility	Computational Role	Example
One	Unit of work	Yes	Yes	Yes — reads from Infinity populations	Center; or- chestration; all work occurs here	Surgeon oper- ating; CPU exe- cut- ing; pilot en- gag- ing; devel- oper de- bug- ging
Infinity	Population await- ing re- duc- tion	Yes (when promoted to tempo- rary One)	Yes (when promoted to temporary One)	No — only self-scoped	Source from which members drawn; passive until promoted	Processes in run queue; pa- tients in wait- ing room; bugs in back- log; radar con- tacts

Table D: Four States

State	Cardinality	Dissolution Status	Processing Entropy	Pipeline Cost	Correct Response	Transition To
Infinity	Infinity	N/A (not yet at One)	Positive (will cost ops when promoted)	Pending (budget reserved for future reduction)	Reduce: enumerate, filter, score, select	One (via reduction)
One	One	Active processing	Positive (being consumed now)	Active (pipeline occupied)	Execute: act, complete, release	Zero-absent (via dissolution over time) or Infinity (release back to population)
Zero-absent	Zero	Dissolved through repetition	Zero (structural processing, no pipeline cost)	Zero (free)	Trust: leave alone; verify dissolution genuine; do not re-introduce management	One (if context change violates validity envelope — cascade)
Zero-external	Zero	N/A (never was at One)	Undefined (processing impossible)	Zero (no processing possible)	Build resilience: measure, approximate, engineer structural responses; accept permanence	No forward transition possible; permanent

Table E: Reduction Pipeline

Stage	Function	Input	Output	Failure Mode	Failure Character	Example (Medical)	Example (Computational)
Enumerate	Make population explicit and finite	Unknown N	Known listable N	Correct answer not in candidate set	Invisible from inside pipeline; most dangerous	Physician doesn't consider correct diagnosis	Scheduler doesn't see ready process
Filter	Eliminate irrelevant candidates	Known N	Smaller relevant N	Over-filter (correct eliminated) or under-filter (too many remain)	Over: correct answer gone; Under: analysis paralysis	Rules out correct diagnosis too early; or fails to narrow differential	Eliminates valid candidate; or scoring overwhelmed by candidates
Score	Evaluate candidates against weighted considerations	Relevant N	Ranked candidate dates	Wrong weights or missing consideration	Correct answer present but outscored	Disease prevalence miscalibrated; key test not ordered	Priority weights wrong; interactivity not considered
Select	Commit highest-scored as One	Ranked candidate dates	One	Cannot commit; oscillation between near-equal candidates	Full information, no action; time budget consumed	Two equally likely diagnoses; physician hesitates	Two equal-priority processes; scheduler oscillates

Table F: Dissolution Examples Across Domains

Domain	First Encounter Element Ops	Competence (R*) Ops	Fully Dissolved Ops	Time to Dissolve	Dissolution Mechanism
Driving	Mirror check 6	1	0	6–12 months	Repetition in consistent driving conditions
Medicine	Classic pneumonia diagnosis 40–60	5–8	0 (pattern fires instantly)	Years of clinical exposure	Thousands of patient encounters
Software	Navigate familiar code-base 25–40	5	0 (fingers go to right file automatically)	Months on same codebase	Daily interaction with code structure
Aviation	Instrument cross-check 6–12	3–4	0 (scan pattern structural)	Months of flight training	Hundreds of hours of instrument practice
Cooking	Knife technique (dice onion) 15–20	5–7	0 (continuous flow)	Weeks to months	Hundreds of repetitions
Language	Native word recognition 3–5 per word	1	0	2–5 years (childhood)	Thousands of encounters per word
Computing	Memory access (cached) ~200 cycles	N/A	~4 cycles (L1 hit)	One access (hardware dissolution)	Cache hardware stores result automatically
Computing	Branch prediction ~15–20 cycles (mis-prediction)	N/A	0 cycles (correct prediction)	Pattern training period	Branch predictor learns from execution history

Domain	Element	First Encounter Ops	Competence (R*) Ops	Fully Dissolved Ops	Time to Dissolve	Dissolution Mechanism
Mathematics	Solving $3x + 7 = 22$	10–15 steps (student)	3 steps	0 (instant recognition: $x = 5$)	Months to years	Hundreds of similar problems
Manufacturing	Assembly operation	$3 \times$ expert time	Minimum physical actions	Continuous flow (near zero conscious ops)	Weeks to months	Repetition with jig support
Music	Scale on instrument	10+ conscious ops per note	1 op per note	0 (fluid performance)	Months to years	Thousands of practice repetitions
Customer support	Known issue resolution	12 ops	2–3	0 (recognized and resolved from pattern)	Weeks to months on support queue	Hundreds of similar tickets

Table G: Cascade Examples

Trigger	Trigger Magnitude	Dissolved Elements Promoted	Cascade Severity	Time Budget	Outcome	Key Insight
Bee in cockpit	Negligible	3 (altitude, heading, attitude)	High	1–2 seconds at cruise	Lane departure likely; performance degradation	Tiny trigger, large cascade

Trigger	Trigger Magnitude	Dissolved Elements Promoted	Cascade Severity	Time Budget	Outcome	Key Insight
Thunder	Large	1 (calm-environment assumption)	Low	Same	Momentary startle, rapid recovery	Large trigger, small cascade
Context switch (CPU)	1 instruction	Hundreds of cache entries	High (hundreds of inflated accesses)	Microseconds	Latency spike; measurable performance hit	Direct cost tiny; cascade cost dominates by $100 \times$
Codebase refactor	Medium	6–15 navigation patterns	High	Days to weeks	60–80% productivity drop for weeks	Developer's dissolved codebase knowledge invalidated
Surgeon announcement (unexpected finding)	Small (one sentence)	3–5 (dissection plan, spatial model, monitoring pattern)	Moderate to high	Anesthesia window	Procedure time extends; risk increases	Information event cascades through dissolved surgical plan

Trigger	Trigger Magnitude	Dissolved Elements Promoted	Cascade Severity	Time Budget	Key Outcome	Insight
Phone notification while driving	Small	2–3 (lane, speed, following distance)	Moderate	0.5–1 seconds	Elevated accident risk for 15–30 seconds	Dissolved driving skills temporarily promoted
Team reorganization	Large	8–15 per team member $\times$ team size	Very high (organizational)	Weeks to months	Sustained productivity drop across organization	Each person's dissolved organizational navigation breaks
Language immersion (new country)	Large (environmental)	20–50 (greetings, requests, reading, conventions)	Very high	Real-time conversation speed	Communication failure for weeks; gradual re-dissolution	Communication dissolved code-book invalidated

Table H: Processing Entropy Comparison

Property	Processing Entropy (H_p)	Shannon Entropy (H)
Measures	Work to reduce element to actionable One	Uncertainty in source; information content
Formula	$H_p(x \mid p, g, c) = \text{op count}$	$H = -\sum p(x) \log_2 p(x)$
Unit	ops	bits
Property of	Relationship: element $\times$ processor $\times$ goal $\times$ context	Source statistics (receiver-independent)
Receiver-dependent?	Yes — same element, different H_p for different processors	No — same H regardless of receiver

Property	Processing Entropy (H_p)	Shannon Entropy (H)
Zero means	Dissolved: processing structural, zero pipeline cost	Deterministic source: no uncertainty
Undefined means	Zero-external: outside processor's domain	N/A (always defined for a source)
Optimization	Reduce to actionability, no further (R^*)	Encode at entropy rate, no fewer bits
Changes with	Processor's dissolution state (decreases with expertise)	Source statistics (fixed for given source)
Maturity trajectory	Decreases toward zero through dissolution	N/A (not a developmental quantity)
Scope	Endpoints (what Shannon excluded)	Channel (what Shannon formalized)

Table I: Concurrency Tax Components

Component	Mechanism	Affects Op Count	Affects Op Duration	Affects Pipeline State	Scales With	Domain Example (Computation)	Domain Example (Human)
Contention	Resource needed by stream is held by another	No	Yes (inflated by wait)	Occupied (waiting)	Resource utilization ρ ; non-linear as $\rho \rightarrow 1$	DB connection pool exhausted; thread waits	Surgeon waits for instrument held by assistant
Cascade	Other stream's (promoted elements cost ops)	Yes	No	Active (re-processing)	Coupling coefficient $\times$ activity rate $\times$ inventory size	Context switch evicts cache; misses spike	Interruption evicts working memory; 15–25 min recovery

Component	Mechanism	Affects Op Count	Affects Op Duration	Affects Pipeline State	Scales With	Domain Example (Computation)	Domain Example (Human)
Coordinating	Multiplexing concurrent access protocols	Yes (coordination ops added)	No	Active (overhead)	Shared resource count $\times$ access frequency $\times$ protocol cost	Mutex lock/unlock; atomic operations	Meeting protocols; status updates; handoff confirmation
Blocking	Pipeline idle waiting for critical resource	No	N/A (no op executing)	Idle (dead time)	Critical resource utilization; competing hold times	Thread sleeping on lock; I/O wait	Waiting for code review; waiting for decision from lead
Interleaving	Deciding which stream to service next	Yes (decision ops added)	No	Active (meta-decision)	Number of ready streams; decision complexity	Scheduler algorithm choosing next process	Developer triaging Slack vs code review vs current bug

Table J: Three-Term Communication Cost

	Hp(A, Scenario encode)	Hs(channel)	Hp(B, decode)	Dominant Term	Shannon Sufficient?	Total Cost Character
Expert → Ex- pert (shared do- main)	~0	Fixed	~0	Channel	Yes	Minimum; Shannon's special case
Expert → Novice	~0	Fixed	High (5–10 ops/undissolved token)	Receiver	No	Receiver overwhelmed; channel perfect
Novice → Ex- pert	High	Fixed	~0	Sender	No	Expert compensates; slow but successful
Novice → Novice	High	Fixed	High	Both end- points	No	Maximum cost; both struggle
Expert → Mixed audi- ence	~0	Fixed	$\sum_i \text{Hp}(B_i)$ — highly varied	Sum of receivers	No	Heterogeneous audience problem
Machine → Ma- chine	~0 (com- piled)	Fixed	~0 (com- piled)	Channel	Yes	Shannon's original model
Teacher → Stu- dent (over les- son)	Moderate (adapt- ing)	Increasing	Decreasing (dissolv- ing)	Shifts over time	Approaches yes as student dissolves	Dynamic optimization
Document → reader popu- lation	One time (author)	Fixed per reader	Varies enor- mously by reader	Σ receiver terms	No	Same document, different cost per reader

Table K: Derivability Classes of R*

Class	R* Status	Knowledge Source	Training Target	Assessment Standard	Example Tasks
P (Prov- able)	Exact value, proven	Task structure (mathematical proof)	Exact: achieve R*	Absolute: distance from proven floor	Sorting (N log N); binary search (log N); parity (N); checklist protocols TSP
B (Bound- able)	Range: R lower $\leq R^* \leq$ R upper	Structural bounds (partial proofs)	Range: achieve R upper, aspire to R lower	Bounded: position within proven range	approxima- tion; numerical integration; criteria- based diagnosis; compressed sensing
E (Em- piri- cal)	Best ob- served; revisable	Expert population (measure- ment)	Relative: match best observed	Percentile: rank in population	General medical diagnosis; tactical combat; creative proof; novel debugging; cooking

Table L: Framework Applications

Application Domain	Primary Framework Contribution	Key Mechanism	Measurable Quantity
System specification	Four flat lists: EntityGroups, EventSets, EventFlows, EventConstraints	Cardinality determines interaction patterns; specification closed under addition	Entry count; completeness by inspection; gap identification
Performance engineering	Fundamental inequality; dissolution as op elimination	Reduce ops, not increase speed; dissolve routine to free budget for novel	Op count; dissolution curve; throughput = $N/(\bar{d} \times \bar{H}_p)$
Training design	Dissolution curves; validity envelopes; derivability classes	Target R^* ; widen envelopes through varied practice; smooth cascade cliffs	Op count over repetitions; envelope width; cascade count under context variation
Expertise assessment	Processing entropy profiles; metric space; skill gap vector	Distance between profiles measures skill difference; direction identifies specific gaps	Profile distance; gap magnitude; gap direction; closure rate
Communication design	Three-term cost; dissolution differential; layered encoding	Minimize total cost (sender ops + bits + receiver ops); add infrastructure where differential is high	Total cost per receiver; dissolution differential per token; audience-weighted efficiency

Application Domain	Primary Framework Contribution	Key Mechanism	Measurable Quantity
Documentation quality	Processing entropy minimization across reader population	Quality = content / total cost; layered encoding for heterogeneous audiences	Reader processing time; lookup rate; completion rate; comprehension score
API design	Processing entropy per consumer; layered surface	Consistent conventions dissolve; progressive disclosure serves heterogeneous consumers	Consumer op count per API call; error rate; time to first successful call
Organizational design	Contention graph; concurrency tax; Brooks's Law derivation	Topology determines tax scaling; optimal team size computable from graph	Tax per team member; marginal tax of additional member; throughput vs team size
Failure prediction	Cascade severity function; fragility profile; dissolution inventory	Predict cascade count from dissolution inventory and validity envelopes before event occurs	Maximum cascade count; cliff locations; plateau coverage
Reliability engineering	Zero-external classification; structural resilience	Classify boundaries correctly; build resilience, not control	Misclassification rate; response infrastructure coverage; cascade recovery capacity

Table M: Complete Equation Reference

Equation	Name	Meaning	Scope
$\Sigma \text{ ops} \times \bar{d} \leq N$	Fundamental inequality	Total processing bounded by time budget; every mechanism in framework reduces left side	Universal; all processors, all domains
$\text{Cost}(A \rightarrow B) = \text{Hp}(A, \text{encode}) + H_s + \text{Hp}(B, \text{decode})$	Three-term communication cost	Total communication cost spans both frameworks; Shannon is middle term	Universal; all communication
$\text{Hp}(x \mid p, g, c)$	Processing entropy	Op count for specific processor, element, goal, context; receiver-dependent	Per element-processor-goal-context tuple
$D(n \mid x, p, \kappa) = C_0 \times f(n, \lambda, \kappa) + R^*$	Dissolution curve	Op count decreasing over repetitions toward zero	Per element-processor pair under consistent context
$S(\Delta c) = \{e : c_0 \in V(e) \wedge (c_0 + \Delta c) \notin V(e)\} $	Cascade severity function	Count of dissolved elements whose envelopes don't contain new context	Per processor's dissolution inventory
$\text{tax}(s, G) = \text{contention} + \text{cascade} + \text{coordination} + \text{blocking} + \text{interleave}$	Concurrency tax equation	Total overhead from execution environment; derivable from contention graph	Per stream in contention graph
$d_2(p, q) = \sqrt{(\sum_i (\text{Hp}(t_i) - \text{Hq}(t_i))^2)}$	Processor distance	Euclidean distance between processing entropy profiles	Per processor pair over shared task set
$g(p, r) = H(p) - H(r)$	Skill gap vector	Per-task cost difference from reference; has magnitude and direction	Per processor-reference pair
$\eta(\text{word}, B) = -\Delta \text{Hp}(B) / \Delta H_s$	Dissolution efficiency	Receiver cost reduction per unit channel cost; guides message optimization	Per word-receiver pair
$\text{Throughput} = N / (\bar{d} \times \bar{\text{Hp}})$	Throughput formula	Units completed per period; increases as average Hp decreases through dissolution	Per processor in operational domain

Table N: Series Map

Paper	Title	Core Contribution	Key Concept Introduced
HOWL-COMP-1 through 10	(Execution pipeline and architecture series)	Runtime architecture; entity system; state machines; utility AI; envelopes	Execution pipeline; envelope mechanism; behavior sets
HOWL-COMP-11	Name Driven Development	Specification method: name every state change before coding	The enum as architecture; the naming act; carrier; action; orchestrator
HOWL-COMP-12	Closed Loop Architecture	Four flat lists specifying complete systems	EntityGroups with cardinality; EventSets; EventFlows; EventConstraints
HOWL-INFO-11	Zero, One, and Infinity	Three cardinalities as intrinsic properties	Permanent/temporary One; interaction patterns; cardinality violations; Zero-to-One transitions
HOWL-INFO-12	Reduction to Cardinality One	Four-stage reduction pipeline	Enumerate→filter→score→select; pipeline cost; pre-computed reductions; OODA mapping
HOWL-INFO-13	Six States / Mathematical Theory / Measurement Theory	Manageability axis; formalization; measurement unit	Six cells; state function; dissolution; processing entropy; the op; fundamental inequality
HOWL-MATH-15	(Measurement Theory)	The op as fundamental unit; fundamental inequality	Op; time budget; throughput formula; dissolution trajectories; concurrency tax definition
HOWL-MATH-16	Geometry of Dissolution and Fragility	Validity envelopes; cascade severity function	Dissolution curve; validity envelope; cascade severity function; cliff/plateau topology; training as envelope engineering
HOWL-MATH-17	Processing Entropy as Metric Space	Expertise has geometry	Processing entropy profile; processor/task distance; matrix factorization; skill gap vector; trajectories

Paper	Title	Core Contribution	Key Concept Introduced
HOWL-MATH-18	Concurrency Tax from System Structure	Overhead derivable from architecture	Contention graph; five tax components; architectural motifs; expert tax discount; Brooks's Law derivation
HOWL-MATH-19	Mathematics of Processing-Aware Communication	Three-term cost equation developed	Compression tokens; dissolution differential; redundancy as infrastructure; heterogeneous audience; layered encoding
HOWL-MATH-20	Derivability Classes of Optimal Reduction	R* classification: provable, boundable, empirical	Three derivability classes; structural properties determining class; hierarchy; classification procedure
HOWL-INFO-14	Bits and Ops	Complete framework in one document	All of the above, in dependency order, self-contained

Table O: Specification Summary

Metric	Count
Core concepts introduced (in dependency order)	30
Equations presented	10
States defined	4 (Infinity, One, Zero-absent, Zero-external)
Cardinalities defined	3 (Zero, One, Infinity)
Reduction pipeline stages	4 (enumerate, filter, score, select)
Concurrency tax components	5 (contention, cascade, coordination, blocking, interleave)
Three-term cost terms	3 (sender ops, channel bits, receiver ops)
Derivability classes	3 (Provable, Boundable, Empirical)
Units defined	2 (bit, op)
Dissolution examples across domains	12
Cascade examples	8
Communication scenarios analyzed	8
Framework applications mapped	10

Metric	Count
Prior papers compressed	20+
Words in main paper	~9,500
Intended reading time	40–60 minutes

HOWL-INFO-14-2026. Bits and Ops: A Complete Theory of Information.

[[HOWL-INFO-14-2026](#)] Bits and Ops. Github: <https://github.com/ghowland/papers/tree/main/papers/INFO/HOWL-INFO-14-2026>

[[HOWL-COMP-1-2026](#)] Implementing a Tall-Infra Data-Only Execution System . Github: <https://github.com/ghowland/papers/tree/main/papers/COMP/HOWL-COMP-1-2026>

[[HOWL-COMP-12-2026](#)] Closed Loop Architecture. Github: <https://github.com/ghowland/papers/tree/main/papers/COMP/HOWL-COMP-12-2026>

[[HOWL-INFO-11-2026](#)] The Relationship of Zero, One, and Infinity in Information Processing. Github: <https://github.com/ghowland/papers/tree/main/papers/INFO/HOWL-INFO-11-2026>

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[[HOWL-MATH-15-2026](#)] A Measurement Theory of Processing. Github: <https://github.com/ghowland/papers/tree/main/papers/MATH/HOWL-MATH-15-2026>

[[HOWL-MATH-20-2026](#)] Derivability Classes of Optimal Reduction. Github: <https://github.com/ghowland/papers/tree/main/papers/MATH/HOWL-MATH-20-2026>